UNIT IV: Wheels, Tyres & Braking System

Syllabus: Wheels and tyres - construction, type & specification, tyre wear and causes; brakes - need, classification, drum & disc, mechanical, hydraulic, pneumatic, vacuum assist, retarders, anti-lock braking system (ABS).

1. Wheels

- The wheel supports the vehicle weight, transmits driving & braking torque to the road through the tyre, and takes steering & cornering forces. It consists of rim, disc (spider), hub.

Requirements

- Strong yet light (low unsprung mass), true running (low radial & lateral run-out), well balanced, good heat dissipation for brakes, corrosion resistant, easy to remove / fit, cheap.

Types of wheels

1. Disc (pressed steel) wheel - rim welded to a stamped steel disc with holes for ventilation; cheap & strong; cars, light trucks.
2. Wire-spoke wheel - steel spokes (laced) between hub & rim; light, strong, good cooling; motorcycles, vintage / sports cars.
3. Cast alloy wheel - Al / Mg alloy cast or forged; light (up to 25 % less), better heat transfer, style; costlier, can crack on impact. Mg alloy -> racing.

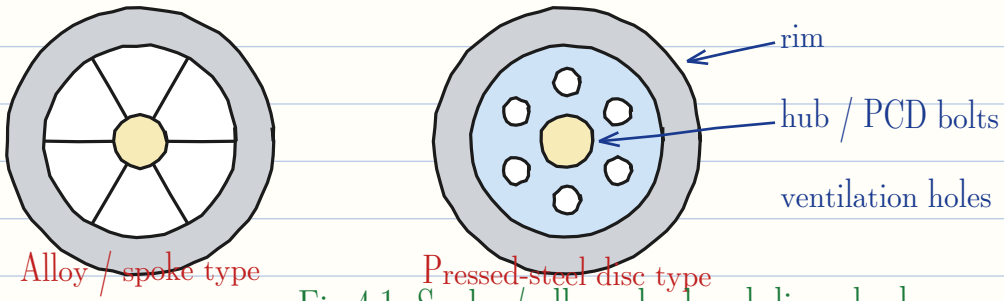


Fig 4.1 Spoke / alloy wheel and disc wheel

Wheel size

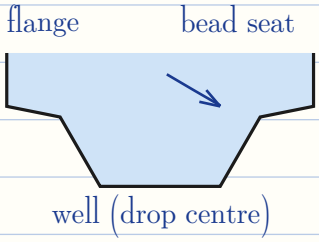
Specified by rim diameter x width, e.g. 15 x 6J (15 inch dia., 6 inch width, J = flange shape), plus PCD (pitch-circle dia. of bolts, e.g. 4 x 100), offset (ET) and centre bore.

2. Rim Construction & Types

- The rim holds the tyre bead; its section is shaped to seat the tyre firmly and (for tubeless) hold air.

Parts of rim

- Flange (side wall support), bead seat (tapered 5 deg - grips bead), well / drop centre (lets the bead drop in during fitting), valve hole, safety hump (tubeless - keeps bead from slipping into well).



Drop-centre rim: most common (cars)
- one piece, well in the middle
- tubeless tyres fit easily

Wide-base / flat-base rim: trucks
- detachable flange, lock ring

Fig 4.2 Drop-centre rim section

Rim types

Type	Feature	Use
Drop-centre (well base)	Deep well in centre for tyre fitting; one piece	Cars, LCVs
Flat-base	Flat seat; demountable flange / lock ring	Trucks, buses
Semi drop-centre	Shallow well	Light trucks
Divided (two-piece)	Halves bolted together	Alloy cars, racing
Multi-piece / split rim	Lock ring, side ring, base	Heavy trucks

Materials

- Steel (pressed from sheet, rolled & welded), Al alloy (cast / forged), Mg alloy. Alloy rims need balancing weights of adhesive type.
- Wheel offset (ET): distance between hub-mounting face & rim centre-line; +ve offset (FWD cars) puts the wheel closer to the arch; affects scrub radius & track.

3. Tyre - Functions & Construction

- A tyre is a flexible toroidal rubber-fabric-steel structure filled with compressed air, mounted on the rim.

Functions

- Carry load (via air pressure); provide traction & braking grip; absorb road shocks (cushion); transmit cornering force; resist wear & heat.

Parts of a tyre

- Tread - thick outer rubber with grooves (pattern) that touches road. Shoulder. Sidewall - protects carcass, carries markings.
- Carcass (casing / plies) - layers of rubberised cord fabric (rayon, nylon, polyester, steel) -> strength & flexibility.
- Belt / breaker - steel / aramid belts below tread (radial) - stiffens tread, resists punctures.
- Bead - bundle of high-tensile steel wire coated with rubber; locks tyre on the rim. Inner liner - air-tight rubber (tubeless). Chafer - protects bead from rim wear.

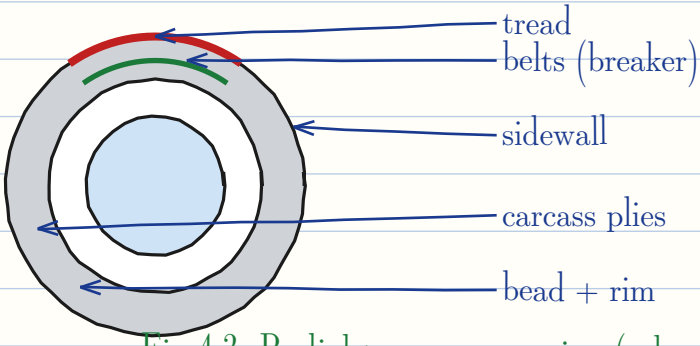


Fig 4.3 Radial tyre cross-section (schematic)

- Tread patterns: rib (highway, low noise), lug (traction / off-road), rib-lug, block / directional, asymmetric; grooves + sipes channel water (prevent aquaplaning).
- Tread compound: natural + synthetic rubber, carbon black, silica, sulphur (vulcanisation), anti-oxidants.

4. Types of Tyres

Based on construction (carcass layout)

1. Cross-ply (bias) tyre - plies at 30-40 deg to the centreline, alternate plies crossed. Stiff sidewall, harsh ride, heat build-up, shorter life; cheap; trucks (older), agri.
2. Belted bias tyre - bias plies + circumferential belts under tread -> better tread life & stability than bias.
3. Radial-ply tyre - plies at 90 deg (radial) to the centreline + steel belts under tread. Flexible sidewall, large contact patch, low rolling resistance, cooler, longer life (30-40 % more), better grip & fuel saving; standard for cars today.

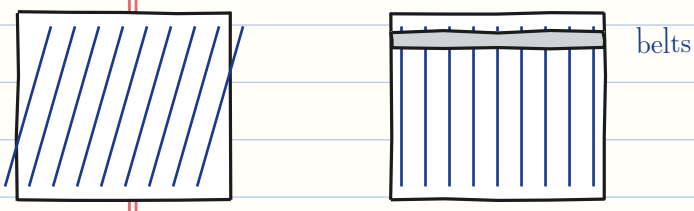


Fig 4.4 Cord direction in cross-ply and radial tyres

Based on air containment

- Tube-type - separate inner tube with valve; puncture -> sudden deflation. Tubeless - inner liner + airtight bead seat, valve on rim; punctures leak slowly, lighter, cooler, safer, easy repair (plug).

Other types

- Run-flat (reinforced sidewall, run 80 km at 80 km/h flat), snow / winter, all-season, off-road (M+S, A/T, M/T), low-profile sports (UHP), space-saver spare.

Point	Cross-ply	Radial
Ride	Harsh	Smooth
Tread life	Shorter	Longer (+30-40 %)
Rolling resistance	High	Low (fuel saving ~3-5 %)
Cost	Low	Higher

5. Tyre Specification & Designation

- Tyre sidewall carries a code describing size, type, load & speed rating. Example: 205 / 55 R 16 91 V.

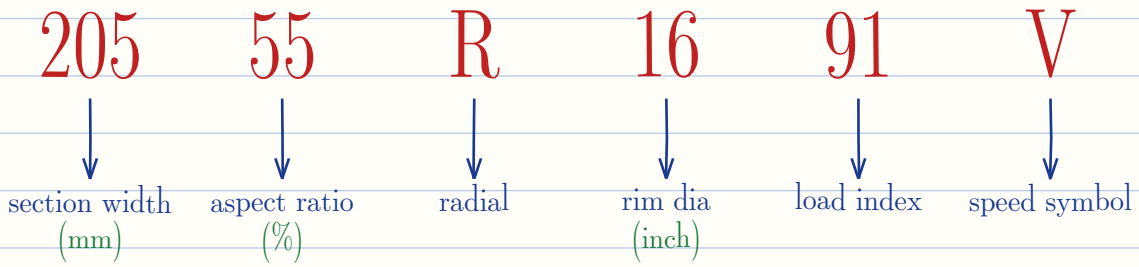


Fig 4.5 Reading a tyre marking

Terms

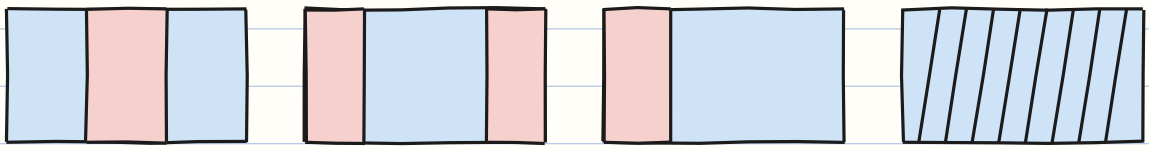
- Section width: widest sidewall-to-sidewall in mm. Aspect ratio (profile): sidewall height / section width x 100 (55 -> sidewall height = 0.55 x 205 = 113 mm); lower = sportier, stiffer.
- R = radial; D = diagonal / bias; ZR = high-speed radial. Rim dia in inches. Old bias format: 6.00 - 16 (width in inch - rim dia).
- Load index (LI): 91 = 615 kg per tyre. Speed symbol: L = 120, N = 140, P = 150, Q = 160, S = 180, T = 190, H = 210, V = 240, W = 270, Y = 300 km/h.

Marking	Meaning
Tubeless / TL	Tubeless tyre
M+S	Mud & snow capability
DOT code	Manufacture plant + week + year (e.g. 2524 = week 25 of 2024)
Treadwear / Traction / Temp	UTQG grades (US): e.g. 400 A A
Max. pressure / load	Printed on sidewall (kPa / psi, kg)

- Recommended inflation pressure: 28 - 36 psi (190 - 250 kPa) for cars; check cold; TPMS warns of low pressure.

6. Tyre Wear & Causes

Tyre wear patterns act as a diagnosis tool - they show what is wrong with inflation, alignment or suspension.



Centre wear — over-inflation | Both-edges wear — under-inflation | One-edge wear — camber | Feathering — toe error

Fig 4.6 Tread-wear patterns (red = worn area)

Wear pattern	Cause	Remedy
Centre worn	Over-inflation	Correct pressure
Both shoulders worn	Under-inflation, overloading	Inflate; reduce load
One shoulder	Excess camber / worn arms	Align; fix suspension
Feathered edge	Wrong toe-in / toe-out	Set toe
Cupping / scalloping	Worn shocks, imbalance	Replace dampers; balance
Flat spots	Locked wheel braking	Use ABS / smooth braking
Diagonal wear	Misalignment, worn bushes	Rotate, align

General causes & prevention

- Hard braking & acceleration, fast cornering, high temperature, rough roads, mixing different tyres, long storage / UV ageing (cracking).
- Tyre rotation every 8,000 - 10,000 km (cross pattern for radials); check pressure monthly; wear indicators (TWI bars at 1.6 mm); replace tread < 1.6 mm; retreading (hot / cold) for trucks.

7. Brakes - Need & Classification

- Brake = device that slows down or stops a vehicle (and holds it stationary) by converting its kinetic energy into heat through friction.

Need / functions

- Reduce speed on demand, stop in shortest distance safely & in a straight line, hold vehicle on slopes (parking), permit higher average speeds.

Requirements

- Effective, progressive (pedal effort proportional to brake force), balanced (front / rear / left / right), minimum fade, quick release, small pedal effort, durable, silent, fail-safe (dual circuit).

Classification

Basis	Types
Purpose	Service (foot) brake, parking (hand) brake, emergency
Method of actuation	Mechanical (rods / cables), hydraulic, pneumatic (air), vacuum-assisted, electric (e-brakes), electromagnetic
Brake unit	Drum (shoe) brake, disc brake, band brake (old transmission brakes)
Location	Wheel brakes, transmission brake (on propeller shaft)
Operation	Foot-operated, hand-operated, automatic (trailer breakaway)
Extra systems	Retarders (exhaust, hydraulic, electric), ABS, EBD

Braking dynamics

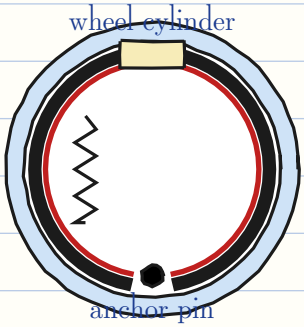
- Retardation a = F / m ; stopping distance s = v^2 / (2 mu g) (v in m/s, mu = tyre-road friction 0.7-0.9 dry). Total = reaction distance + braking distance.
- Weight transfer to the front during braking -> front brakes do ~ 70 % of work -> bigger front discs; rear brake force limited by a proportioning valve to avoid rear lock-up.

Brake fade

Loss of braking effectiveness caused by overheating of linings / fluid (vapour). Disc brakes fade less than drum brakes.

8. Drum Brakes - Construction

- A rotating drum (bolted to the wheel hub) is pressed from the inside by two curved brake shoes with friction linings, mounted on a stationary back plate (backing plate).



Brake drum (cast iron, rotates)
Brake shoes + linings
Return (retracting) springs
Wheel cylinder (hydraulic) / cam
Back plate + anchor pin
Self-adjuster / hand-brake lever

Fig 4.7 Drum brake (leading-trailing type)

Working

- Pedal pressure -> wheel cylinder pistons push shoes outward against the rotating drum -> friction slows the wheel. Release -> return springs pull shoes back.

Materials

- Drum: grey cast iron (sometimes with steel liner / Al fins); Shoes: pressed steel / Al; Lining: organic (non-asbestos), semi-metallic, ceramic - riveted / bonded; friction coefficient 0.35 - 0.45.

Self-energising (servo) action

- When the drum rotation drags the shoe in the direction of rotation ("leading shoe "), friction wedges it further against the drum -> extra braking force without more pedal effort. Trailing shoe is pushed away -> no self-energisation.
- Disadvantages: heat retained inside drum -> fade, water retention, adjustment needed; advantages: cheap, strong parking brake, protected from dirt.

9. Drum Brake Types & Mechanical Brakes

Type	Arrangement	Remark
Leading-trailing	One wheel cylinder; one leading + one trailing shoe (same anchor)	Equal in both directions; rear brakes of cars
Two leading shoe	Two single-piston cylinders; both shoes are leading in forward motion	Strong forward braking, poor in reverse; front brakes (older)
Duo-servo	Both shoes linked (floating); primary shoe pushes secondary	Highest self-energisation; US cars, trucks
Single anchor / Twin leading	Two anchors (two pivots)	Motorcycles, scooters (mechanical)

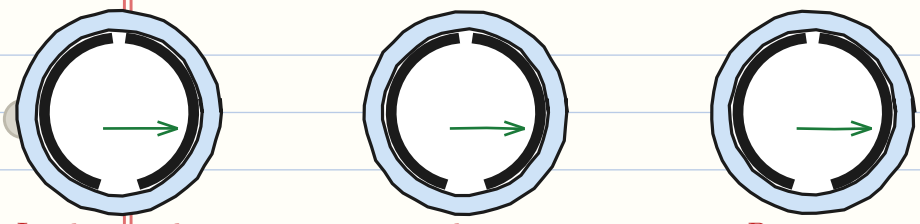


Fig 4.8 Drum brake arrangements

Mechanical brake linkages

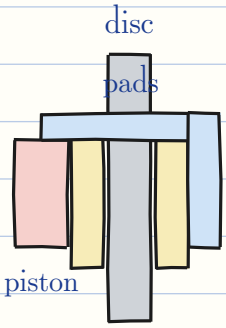
- Pedal / hand lever -> rods / cables / levers / cam or wedge -> shoes. Equaliser makes the force same at each wheel.
- Used for parking (hand) brake even in hydraulic cars: cable -> lever in rear shoe -> expands shoes (duo-servo / drum-in-disc). Also motorcycle rear brake, old cars.
- Disadvantages: friction & lost motion in linkage, unequal braking, frequent adjustment, hard to apply large force.
- Cam-operated (S-cam) brakes in air brakes: S-cam spreads shoes, turned by slack adjuster.

Brake adjustment

Self-adjusters (ratchet / star-wheel) maintain shoe-to-drum clearance 0.2 - 0.4 mm; manual adjust where no auto adjuster; slack adjuster in air brakes.

10. Disc Brakes

- A flat disc (rotor) is bolted to the wheel hub; a caliper straddles it holding two friction pads that squeeze (clamp) the disc from both sides when hydraulic pressure acts on the caliper piston(s).



Disc / rotor (cast iron, vented)
Brake pads (friction material)
Caliper (piston + seals)
Hydraulic inlet / bleed screw
Dust boot, retaining pins

Fig 4.9 Disc brake assembly (schematic)

Types of caliper

- Fixed caliper: rigid, piston(s) on both sides (2 - 8 pots); better stiffness, more costly - sports/heavy cars.
- Floating (sliding) caliper: pistons on inboard side only; caliper slides on guide pins to apply the outer pad; cheap, compact - most cars.

Disc types

- Solid, ventilated (internal fins), drilled / slotted (gas & water clearing), carbon-ceramic (racing). Pad: semi-metallic, ceramic; wear sensor / squealer.

Point	Disc	Drum
Heat dissipation	Excellent (open)	Poor (enclosed)
Fade / wet performance	Better	Worse
Self-energisation	None (needs servo)	Yes
Weight / adjust	Light, self-adjusting	Heavier, manual adj.
Use	Front (all), rear on cars	Rear (economy), trucks

11. Hydraulic Brake System

- Based on Pascal's law - pressure applied to a confined fluid is transmitted equally in all directions. Pedal force -> master cylinder -> fluid pressure (up to 100-150 bar) -> wheel cylinders / calipers.

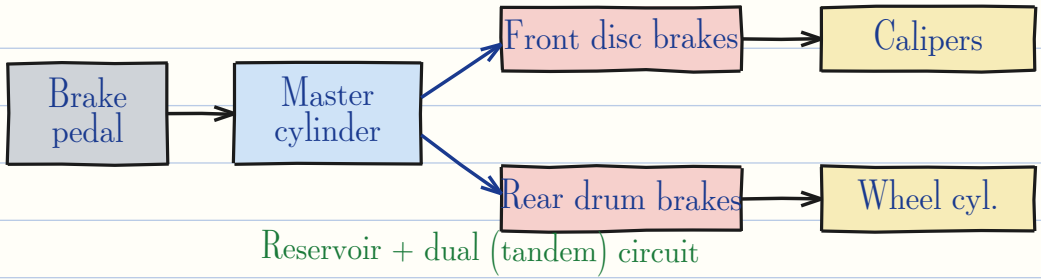


Fig 4.10 Hydraulic brake layout (diagonal / front-rear split)

Components

- Brake pedal (lever ratio 4:1 - 6:1), master cylinder (tandem) + reservoir, brake lines (steel pipes, flexible hoses), wheel cylinders (drum) / caliper pistons (disc), proportioning / pressure-limiting valve, brake switch.

Working

1. Pressing pedal pushes master piston -> fluid pressure rises in the line.
2. Same pressure reaches all wheel pistons; force at wheel = pressure x piston area -> shoes / pads apply.
3. On release, return springs push pistons back; fluid returns to master cylinder (residual pressure valve ~ 0.5 bar keeps cups sealed).

- Advantages: equal pressure to all wheels, fast action, less friction, easy layout, self-lubricating, high force multiplication. Disadvantages: air in system (spongy pedal), leak = complete loss (so dual circuit), fluid absorbs moisture.

12. Master Cylinder, Fluid & Bleeding

Tandem (dual-circuit) master cylinder

- Two pistons in one bore - primary (pushrod-operated) and secondary (floating) - each feeds its own circuit (front / rear or diagonal). If one circuit leaks, the other still brakes (legal requirement).

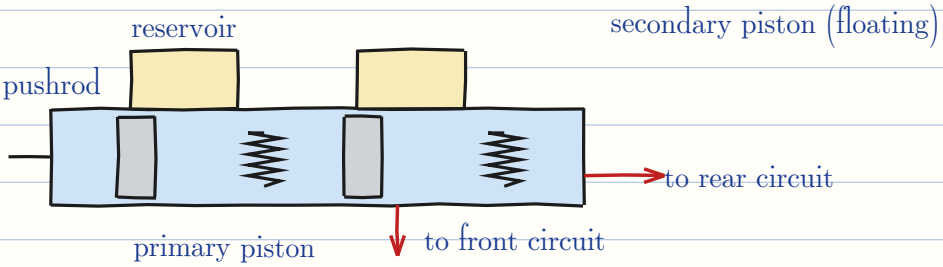


Fig 4.11 Tandem master cylinder

Brake fluid

- Glycol-ether based (DOT 3, DOT 4, DOT 5.1); DOT 5 = silicone (do not mix). Properties: high boiling point (DOT 3: 205 deg C dry; DOT 4: 230 deg C), low freezing point, no corrosion, compatible with rubber seals, good lubricity.
- Hygroscopic - absorbs moisture -> lowers boiling point -> vapour lock / fade; change fluid every 2 years.

Bleeding of brakes

- Removes air (compressible) from lines: open bleed screw while pedal is pressed (manual / pressure / vacuum bleeding) in order farthest wheel from master cylinder first; keep reservoir topped up.

Fault	Likely cause
Spongy pedal	Air in lines; low fluid
Pedal sinks slowly	Master cylinder cups leaking
Pulling to one side	Stuck caliper / contaminated lining
Dragging brakes	Seized piston, faulty return spring
Squeal / judder	Glazed pads, warped disc

13. Pneumatic (Air) Brakes

- Use compressed air (6 - 8 bar) to apply brakes - standard on heavy trucks, buses, trailers (high force, light pedal effort, long vehicles / trailers coupling).

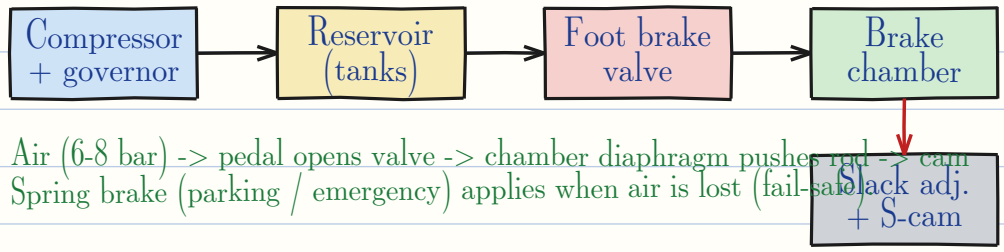


Fig 4.12 Air brake layout (simplified)

Main components

- Compressor (engine-driven), governor (cut-in 6.5 / cut-out 8.5 bar), air dryer, reservoirs (wet / primary / secondary), foot brake valve (treadle), relay valve (quick action at rear), brake chamber (diaphragm), slack adjuster, S-cam, quick release valve, pressure protection valve, trailer gladhands (couplings).

Working

- Pedal -> treadle valve admits air from tank to brake chambers -> push rod -> slack adjuster -> S-cam -> shoes press drum. Release -> air exhausts through quick release valve; return springs retract shoes.
- Spring brakes (park / emergency): powerful spring applies the brake when air pressure falls below ~ 4 bar.

Point	Advantage	Disadvantage
Air brake	Unlimited air source, light pedal, fail-safe, trailers easy	Delay in response (lag), bulky, costly
Hydraulic	Fast, compact	Limited force, leaks

14. Vacuum Assisted Brakes (Servo)

- A vacuum servo (booster) multiplies the driver's pedal force 2-4 times using engine intake manifold vacuum (diesel: vacuum pump) acting on a large diaphragm. Used with hydraulic brakes in cars & LCVs.

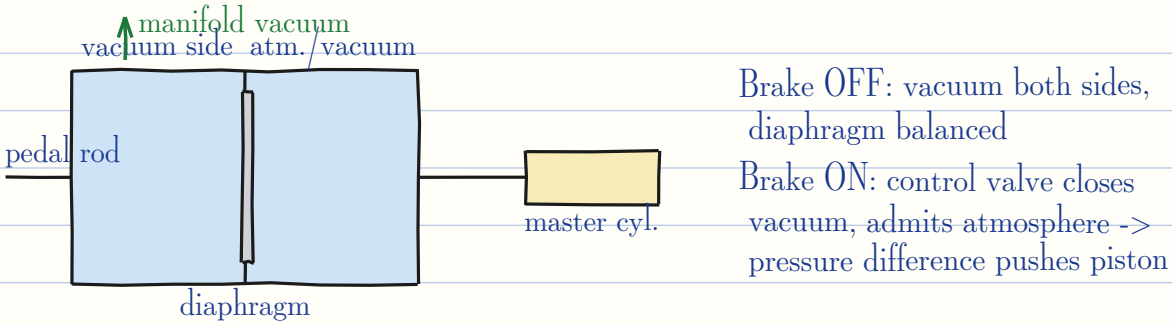


Fig 4.13 Vacuum servo (brake booster)

Working

1. Released: both sides of diaphragm connected to manifold vacuum -> no net force.
2. Pedal pressed: pedal rod moves the valve plunger -> vacuum port closed, atmospheric air admitted to the rear chamber.
3. Pressure difference (atm - vacuum = ~0.7 bar) x diaphragm area gives extra thrust to the master cylinder pushrod.
4. Fail-safe: if vacuum fails, the pedal rod still acts mechanically on the master cylinder (needs higher effort); a check valve retains vacuum for 2-3 assisted brakes.

Types

- Direct-acting booster (on firewall, most cars), tandem booster (two diaphragms, trucks), hydro-vac (hydraulic booster), electric vacuum pump for diesel / EV, brake-by-wire (iBooster).

15. Retarders

- A retarder is an auxiliary braking device that absorbs energy continuously without friction brakes - controls speed on long descents, prevents brake fade, saves brake wear (40-50 %). Common on heavy trucks, buses, hill-area vehicles.

Types of retarders

1. Exhaust brake: butterfly valve in the exhaust pipe closes -> back pressure on the exhaust stroke -> engine acts as a compressor and slows down. Simple, cheap (~ 30-50 % of engine power).
2. Engine (compression release / Jake) brake: exhaust valves are opened near TDC at the end of compression -> compressed air released (not returned to the piston) -> strong braking (up to engine rated power); noisy.
3. Hydraulic retarder (hydrodynamic): a rotor on the driveshaft churns oil against a fixed stator inside a housing; oil friction produces braking torque, heat removed by the engine cooling circuit; torque proportional to speed^2 (Voith, ZF).
4. Electric / eddy-current retarder: steel rotor discs rotate between electromagnet coils on the frame; eddy currents in discs resist rotation -> braking, heat dissipated by air-cooled discs (Telma). No contact, no wear; stages by coil current.

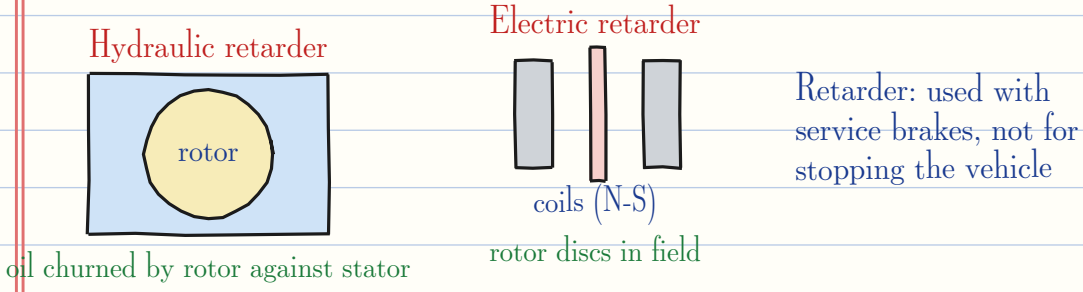


Fig 4.14 Hydraulic and electric retarders

- Retarders reduce speed but cannot hold a vehicle at rest.

16. Anti-lock Braking System (ABS)

- ABS prevents the wheels from locking during hard braking by automatically modulating brake pressure many times a second -> vehicle keeps steering control and shortest possible stopping distance on most surfaces.

Need

- A locked wheel skids (sliding friction < peak friction), loses cornering grip (cannot steer), wears flat spots on tyres. Maximum braking happens at 10 - 20 % slip.

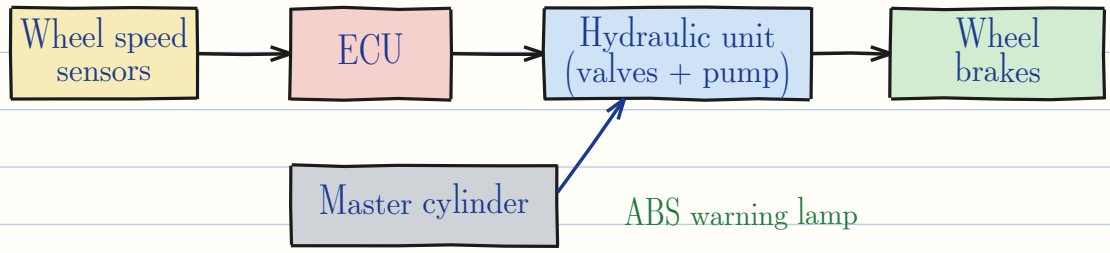


Fig 4.15 ABS block diagram

Components

- Wheel speed sensors (inductive / Hall) + toothed tone ring; ECU; hydraulic modulator - inlet (normally open) & outlet (normally closed) solenoid valves, accumulator & return pump; warning lamp.

Working cycle (3 phases)

1. Pressure build / hold: normal braking; if sensor shows a wheel decelerating too fast (about to lock), ECU closes inlet valve - pressure is held.
2. Pressure release: ECU opens outlet valve; fluid goes to accumulator, pump returns to master cylinder -> wheel speeds up.
3. Pressure re-apply: inlet valve reopens; the cycle repeats up to 15-20 times / s (pedal pulsates).

- Channels: 4-channel 4-sensor (best), 3-channel, 1-channel (rear, pickups). Two-wheeler ABS: 1 / 2 channel.

17. ABS Advantages & Brake Troubleshooting

Advantages of ABS

- Steering control kept under panic stops; shorter stopping distances on wet / slippery roads (on loose gravel / snow may be longer); no tyre flat-spotting; stable vehicle; basis for EBD, TCS, ESP, Brake Assist.

Limitations

- Costly; does not defy physics (speed, bad tyres); pedal pulsation startles drivers (do not pump - press hard and hold); may lengthen stopping distance on soft surfaces.

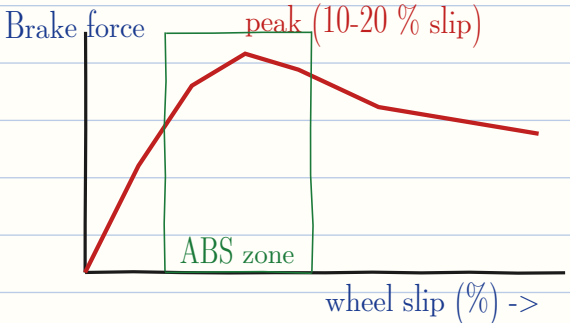


Fig 4.16 Friction vs wheel slip

Brake system troubleshooting

Symptom	Cause	Remedy
Weak brakes	Worn linings, low fluid, air, faulty servo	Replace, bleed, check booster
Brake pull	Uneven pads, tyre pressure, stuck caliper	Service caliper / equalise
Squealing	Glazed / worn pads, no shim	Replace pads, grease backs
Vibration	Warped disc / drum out-of-round	Resurface or replace

Servicing hint

Replace pads in axle pairs; check disc thickness (min mark), bleed in correct order, never reuse brake fluid, and test-drive after service.